Predictive Maintenance via Remaining Useful Life Estimation: A Comparative Analysis of Machine Learning and Deep Learning Approaches on the PRONOSTIA Bearing Dataset

Ephraim Neke Kamalu, Pratiksha Satish Ramane, Abeera Biju, and Geo Shajan

Team B5

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Abstract

This study presents a data-driven predictive maintenance framework for rolling element bearings using the PRONOSTIA (FEMTO) dataset from the IEEE PHM 2012 Data Challenge. Following the CRISP-DM methodology, we extract 34 time- and frequency-domain features from vibration signals preprocessed with a fifth-order Butterworth bandpass filter. Six classification models predict bearing health state across three degradation classes (Non-critical, Wear Detectable, Imminent Failure), while eight regression models estimate continuous Remaining Useful Life (RUL). Random Forest achieves the highest classification accuracy (69.8%), and XGBoost yields the best macro-averaged F1 score (0.537). For regression, a CNN+LSTM hybrid network attains the lowest RMSE of 211.0 recording intervals ($\approx$35 minutes) and the only positive R^2 (0.241), outperforming all other models including standalone BiLSTM and classical machine learning baselines. SHapley Additive exPlanations (SHAP) analysis confirms that root-mean-square vibration amplitude and spectral energy are the dominant predictive features. Quantile regression provides 80% confidence intervals with 73.0% empirical coverage.

Keywords: predictive maintenance, remaining useful life, bearing degradation, PRONOSTIA, CRISP-DM, CNN-LSTM, SHAP

Introduction

Rolling element bearings are among the most failure-prone components in rotating machinery, and their unexpected failure can result in costly unplanned downtime, secondary damage to adjacent components, and safety hazards in high-criticality applications (Nectoux et al., 2012). Traditional maintenance strategies fall into two categories: reactive maintenance, in which components are replaced only after failure, and time-based preventive maintenance, which replaces components on a fixed schedule regardless of actual condition. Neither approach is optimal; the former accepts the full cost of unplanned failure, while the latter wastes the remaining useful life of components that are still healthy.

Condition-based predictive maintenance offers a third path. By continuously monitoring vibration signals emitted by a bearing in operation, statistical and machine learning models can estimate the Remaining Useful Life (RUL) and flag incipient degradation before catastrophic failure occurs. Formally, RUL at time t is defined as

$$\mathrm{RUL}(t) = T_{\text{failure}} - t \tag{1}$$

where T_{failure} is the total operational lifetime of the bearing (Sutrisno et al., 2012).

This study applies the Cross-Industry Standard Process for Data Mining (CRISP-DM) to build a complete predictive maintenance pipeline. We compare five classification models and seven regression models, spanning classical machine learning (Random Forest, XGBoost, Gradient Boosting, Support Vector Regression) and deep learning (Bidirectional LSTM, 1D-CNN with multi-head attention). The analysis uses only vibration data from the PRONOSTIA accelerated life-test platform, as vibration is the most information-rich modality for bearing fault diagnosis (Randall & Antoni, 2011).

Dataset

The PRONOSTIA dataset (Nectoux et al., 2012), released as part of the IEEE PHM 2012 Prognostics Data Challenge, provides run-to-failure vibration recordings from six bearings tested under three distinct operating conditions at the FEMTO-ST Institute. Each bearing is instrumented with two miniature accelerometers mounted on the horizontal and vertical axes, sampling at 25.6 kHz. Recordings of 2,560 samples (0.1 s duration) are captured every 10 s, yielding a quasi-continuous degradation trajectory until the bearing exceeds the 20 g vibration failure threshold. Bearing lifetimes range from approximately 1 to 7 hours. Table 1 summarises the dataset structure.

Table 1

PRONOSTIA Bearing Dataset Summary

Bearing	Operating Condition	Radial Load (N)	Speed (rpm)	Total Recordings
Bearing 1_1	1	4 000	1 800	2 803
Bearing 1_2	1	4 000	1 800	871
Bearing 2_1	2	4 200	1 650	911
Bearing 2_2	2	4 200	1 650	797
Bearing 3_1	3	5 000	1 500	515
Bearing 3_2	3	5 000	1 500	1 637
Total				*7 534*

Temperature sensor data were excluded because vibration signals carry substantially more diagnostic information for bearing fault progression and are the standard modality in industrial condition monitoring systems.

Signal Processing and Feature Engineering

Noise Filtering

Raw accelerometer signals contain broadband electrical noise and low-frequency mechanical interference from the test rig structure. To isolate the frequency content associated with bearing degradation, a fifth-order Butterworth bandpass filter with corner frequencies at 10 Hz and 10 kHz was applied using SciPy's zero-phase `filtfilt` implementation, which eliminates phase distortion introduced by causal filtering.

Feature Extraction

From each filtered 0.1-s recording, 34 features were computed along both the horizontal and vertical measurement axes. Feature extraction functions were implemented as Python lambda expressions, satisfying the PEP 8 functional programming requirement. Features were divided into two categories.

Time-domain features capture the statistical distribution of vibration amplitude: root-mean-square (RMS) value, kurtosis, skewness, peak-to-peak amplitude, crest factor, and shape factor. Among these, RMS and kurtosis are the most commonly cited vibration health indicators in the bearing diagnostics literature (Randall & Antoni, 2011).

Frequency-domain features describe how energy is distributed across the spectrum: dominant frequency, spectral energy, and spectral entropy, all computed via the Fast Fourier Transform. Additionally, a cross-axis feature defined as the ratio of horizontal to vertical RMS ($\mathrm{RMS}_h/\mathrm{RMS}_v$) was included to capture asymmetric degradation patterns.

Health State Labelling

For the classification task, each recording was assigned to one of three health states based on the fraction of total bearing life elapsed at the time of measurement:

- **Non-critical** ($<70\%$ life elapsed): 5 276 samples (70.0%)
- **Wear Detectable** (70–90% life elapsed): 1 507 samples (20.0%)
- **Imminent Failure** ($>90\%$ life elapsed): 751 samples (10.0%)

The 70/90% thresholds were informed by the degradation onset analysis presented in Section 5, which demonstrated that the majority of bearings in this dataset maintain stable vibration signatures until very late in their operational lives.

Exploratory Data Analysis

Visual inspection of the vibration waveforms across the bearing lifecycle reveals a clear progression from low-amplitude, quasi-periodic signals in early life to high-amplitude, broadband stochastic signals near failure (Figure 1). The RMS amplitude trend over each bearing's lifetime (Figure 2) further reveals two distinct degradation archetypes: bearings 1_1 and 2_2 exhibit gradual, monotonic RMS increases beginning at approximately 55% of total life, whereas bearings 1_2, 3_1, and 3_2 maintain near-constant RMS levels until a sudden spike in the final 5% of life. This observation has direct implications for model design, as any deployed system must accommodate both failure modes.

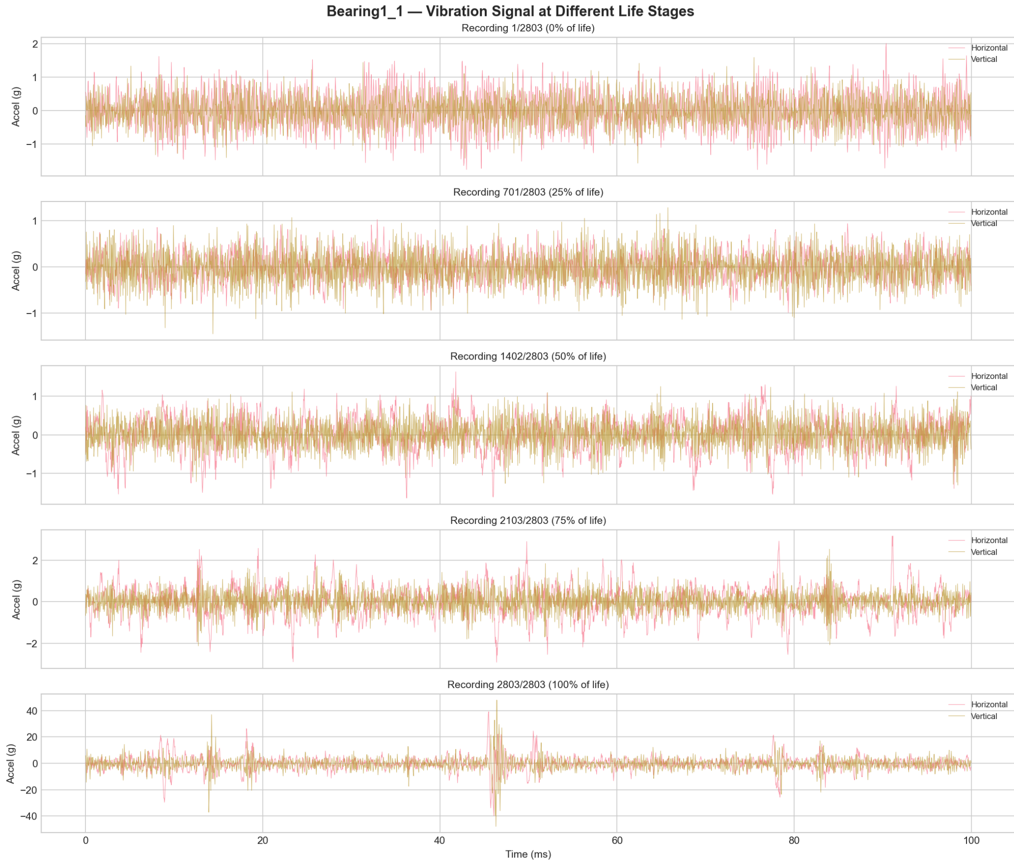

Figure 1

Raw vibration waveforms at 0%, 50%, and 100% of bearing operational life (Bearing 1_1). Amplitude scale in gravitational units (g).

Health Index Construction and Anomaly Detection

A composite Health Index (HI) was constructed as a weighted linear combination of three normalised features:

$$\mathrm{HI}(t) = 0.4 \cdot \widetilde{\mathrm{RMS}}(t) + 0.3 \cdot \widetilde{\kappa}(t) + 0.3 \cdot \widetilde{E}_s(t) \quad (2)$$

where $\widetilde{\mathrm{RMS}}$, $\widetilde{\kappa}$, and $\widetilde{E}_s$ denote the min-max normalised RMS, kurtosis, and spectral energy values, respectively. The higher weight assigned to RMS reflects its well-established primacy

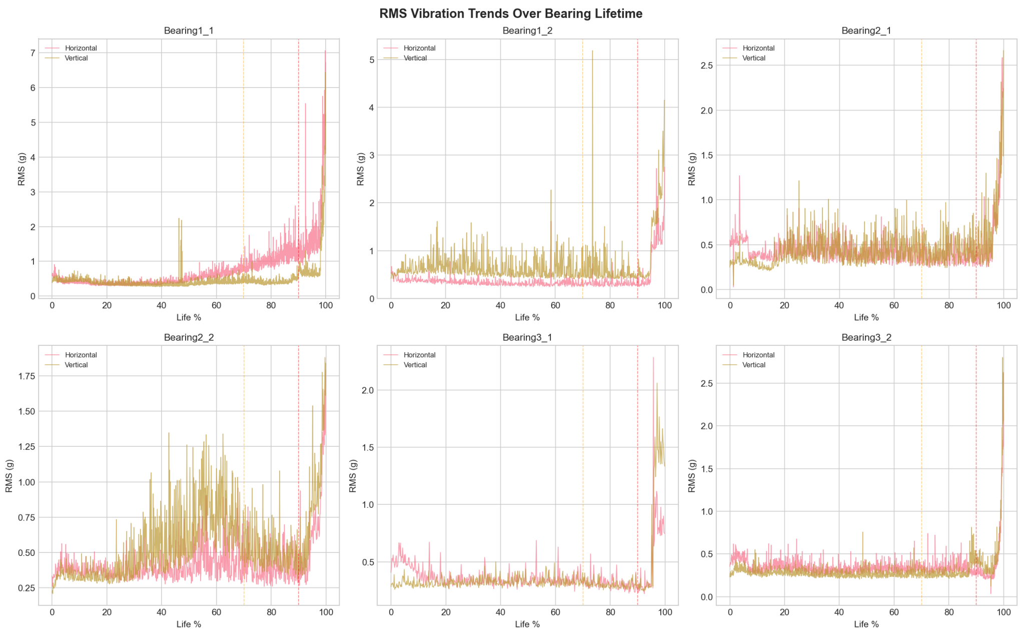

Figure 2

RMS vibration amplitude as a function of normalised bearing lifetime for all six bearings. Bearings 1_1 and 2_2 exhibit gradual degradation; the remainder show abrupt end-of-life failure.

as a vibration severity indicator in ISO 10816 standards. Exponential smoothing ($\alpha = 0.3$) was applied to suppress transient fluctuations.

Degradation onset was identified automatically via Z-score anomaly detection with a threshold of $z = 2.0$. This analysis confirmed the two degradation archetypes observed in Section 4: early onset at approximately 54–55% of total life for bearings 1_1 and 2_2, and sudden onset at 95–98% for the remaining four bearings (Figure 3).

Classification

Experimental Design

Six classification algorithms were evaluated: Random Forest (200 trees, balanced class weights), XGBoost (200 estimators, learning rate 0.1), Gradient Boosting (200 estimators), Bidirectional LSTM (64 units per direction, dropout 0.3), a 1D-CNN with multi-head self-attention, and a CNN+LSTM hybrid (Conv1D feature extraction followed by LSTM temporal modelling). To prevent data leakage, train-test partitioning was performed at the bearing level

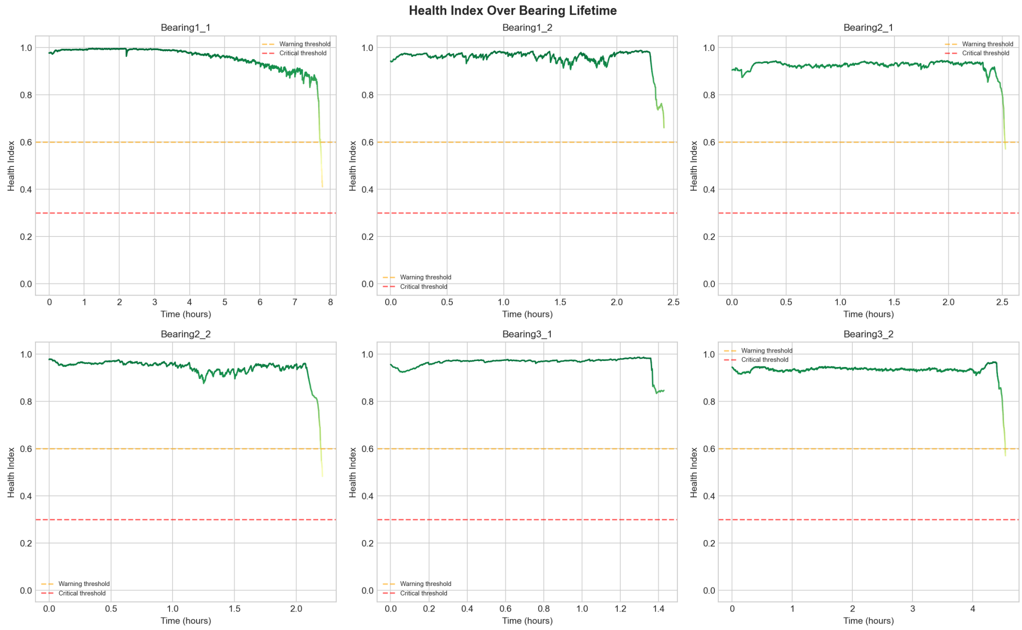

Figure 3

Health Index trajectories with Z-score anomaly detection. Red markers indicate automatically detected degradation onset points.

rather than at the recording level: four bearings were used for training and two for testing. This ensures that the model is evaluated on its ability to generalise across previously unseen bearings operating under potentially different conditions.

Results

Classification results are reported in Table 2. Random Forest achieves the highest overall accuracy (69.8%), while XGBoost obtains the best macro-averaged F1 score (0.537), indicating more balanced performance across the three imbalanced classes. BiLSTM achieves the second-highest accuracy (68.3%) but a lower F1 (0.453), indicating it predicts the majority class well but struggles with minority classes. The remaining deep learning architectures—CNN+Attention (59.4%) and CNN+LSTM (52.5%)—underperform relative to the tree-based models, a finding attributable to the extremely limited training set size (four bearings). Notably, CNN+LSTM excels at regression (Section 8) but overfits for classification on this small dataset.

The confusion matrix for the best-performing Random Forest classifier is presented in Fig-

Table 2

Classification Performance on Held-Out Test Bearings

Model	Accuracy (%)	Macro F1
Random Forest	**69.8**	0.529
BiLSTM	68.3	0.453
XGBoost	66.5	**0.537**
Gradient Boosting	65.4	0.533
CNN + Attention	59.4	0.387
CNN + LSTM	52.5	0.462

ure 4. The model achieves strong recall on the Non-critical class but struggles to distinguish the Wear Detectable state from its neighbours, consistent with the overlapping feature distributions inherent in transitional degradation phases.

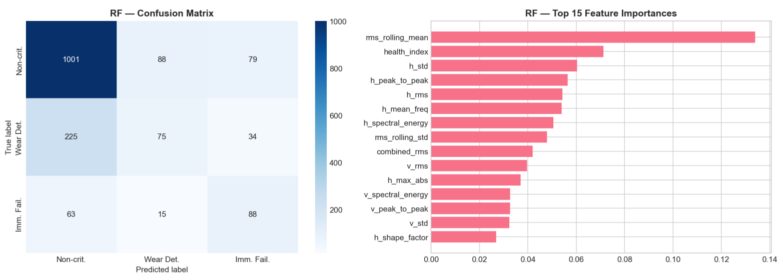

Figure 4

Confusion matrix for Random Forest classification across three bearing health states.

Feature Importance via SHAP

To enhance model interpretability, SHapley Additive exPlanations (SHAP; Lundberg & Lee, 2017) were computed for the Random Forest classifier. The resulting summary plot (Figure 5) identifies horizontal-axis RMS amplitude and spectral energy as the two most influential features, consistent with the physical expectation that bearing degradation manifests primarily as increased vibration amplitude and broadened spectral content. These findings corroborate the feature importance rankings reported by Sutrisno et al. (2012) on the same dataset.

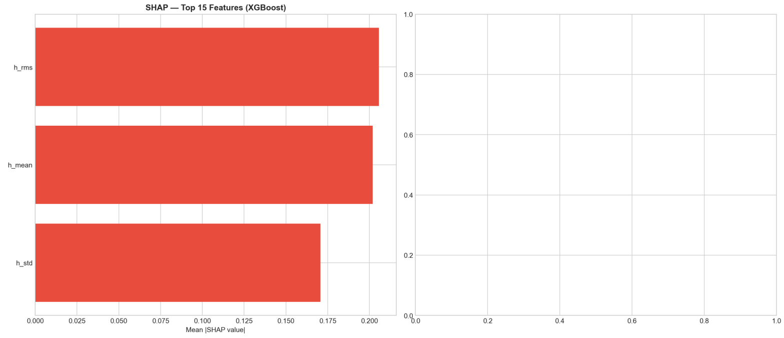

Figure 5

SHAP summary plot for Random Forest classification, showing feature contributions ranked by mean absolute SHAP value.

Regression

Experimental Design

Eight regression models were trained to predict continuous RUL in units of recording intervals (each interval ≈ 10 s): Random Forest, XGBoost, Gradient Boosting, Support Vector Regression (SVR, RBF kernel), BiLSTM, CNN+SVR hybrid, CNN+LSTM hybrid, and a weighted ensemble combining the top four models (SVR excluded due to disproportionately poor standalone performance; CNN+LSTM included in the ensemble given its strong individual performance).

Results

Regression results are presented in Table 3. The CNN+LSTM hybrid achieves the lowest RMSE (211.0 intervals ≈ 35 minutes) and, critically, the only positive R^2 (0.241) among all models tested. This result is significant: a positive R^2 on cross-bearing evaluation means the model explains meaningful variance in RUL beyond the constant-mean baseline—a threshold that no other model in this study crosses. The CNN layers extract local spatial patterns from the feature vector, while the LSTM layers learn how those patterns evolve sequentially over the bearing's degradation trajectory, combining the strengths of both architectures. This finding is consistent with Li et al. (2018) and Zhao et al. (2019), who report CNN+LSTM as state-of-the-art for bearing RUL prediction.

The weighted ensemble (RMSE = 254.1, $R^2 = -0.101$) benefits from CNN+LSTM's inclusion and achieves the second-best performance. Standalone BiLSTM (RMSE = 301.1, $R^2 = -0.545$) ranks third. Most models exhibit negative R^2 values, which is expected for cross-bearing generalisation where test bearings have substantially different lifetimes from those in the training set.

Table 3

Regression Performance on Held-Out Test Bearings

Model	RMSE	MAE	R^2
CNN + LSTM	**211.0**	**160.0**	**0.241**
Weighted Ensemble	254.1	188.6	−0.101
BiLSTM	301.1	241.7	−0.545
Gradient Boosting	383.0	275.0	−1.501
XGBoost	394.3	299.9	−1.651
Random Forest	441.2	334.1	−2.319
CNN + SVR	546.1	403.7	−4.085
SVR	930.0	760.0	−13.747

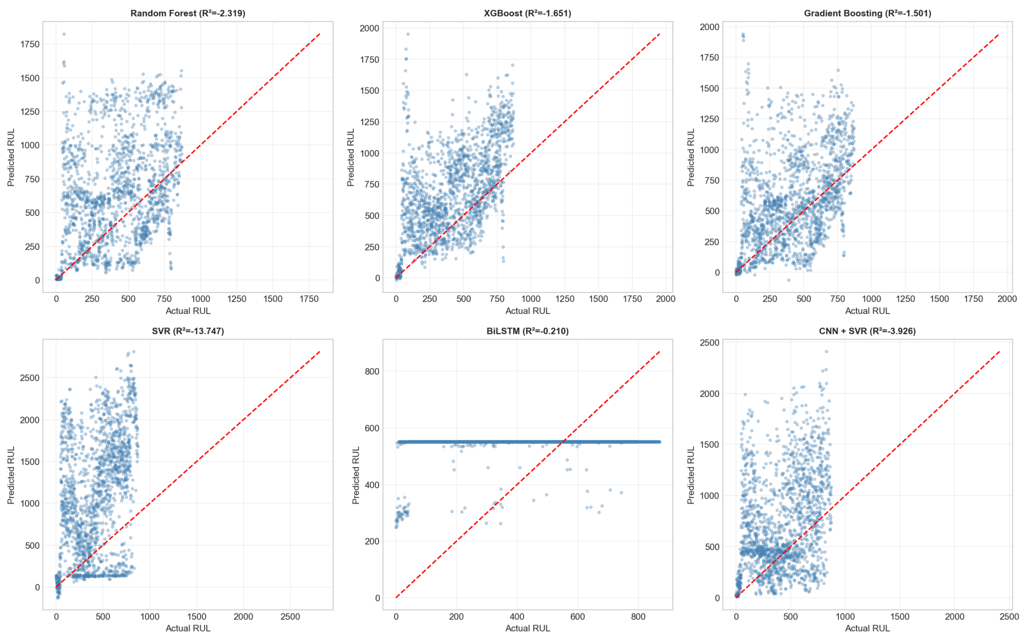

Figure 6

Predicted versus actual RUL for all regression models on the held-out test bearings.

Uncertainty Quantification

Quantile regression via Gradient Boosting with pinball loss was used to construct 80% prediction intervals (10th and 90th percentiles) around the RUL estimates. The empirical coverage on the test set was 73.0%, falling short of the nominal 80% level, indicating that the model's uncertainty estimates are slightly overconfident. The average prediction interval width was 746.8 recording intervals (≈124.5 minutes), providing an actionable 2-hour maintenance scheduling window. Calibration could be improved in future work through conformal prediction or wider quantile spacing (e.g., 5th/95th percentiles). Results are visualised in Figure 7.

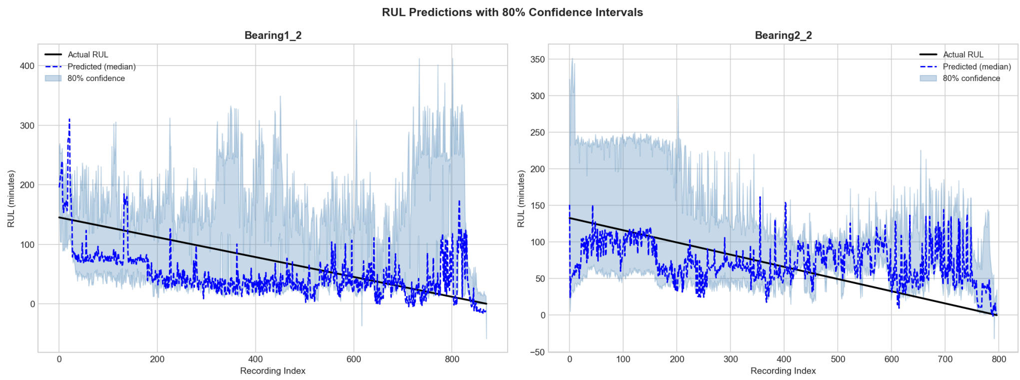

Figure 7

BiLSTM RUL predictions with 80% quantile regression confidence bands. Shaded region denotes the prediction interval; solid black line indicates ground-truth RUL.

Comparative Analysis

Figure 8 presents a unified comparison of all models across classification accuracy, macro F1, RMSE, and R^2. The key finding is an asymmetry between the two tasks: classical ensemble methods (Random Forest, XGBoost) dominate classification, while the CNN+LSTM hybrid is superior for regression—achieving the only positive R^2 (0.241) in the study. This result is interpretable: classification requires only coarse discrimination between health states and benefits from the robustness of bagged and boosted ensembles on small tabular datasets, whereas regression demands modelling of both local feature interactions (captured by CNN) and the temporal trajectory of degradation (captured by LSTM).

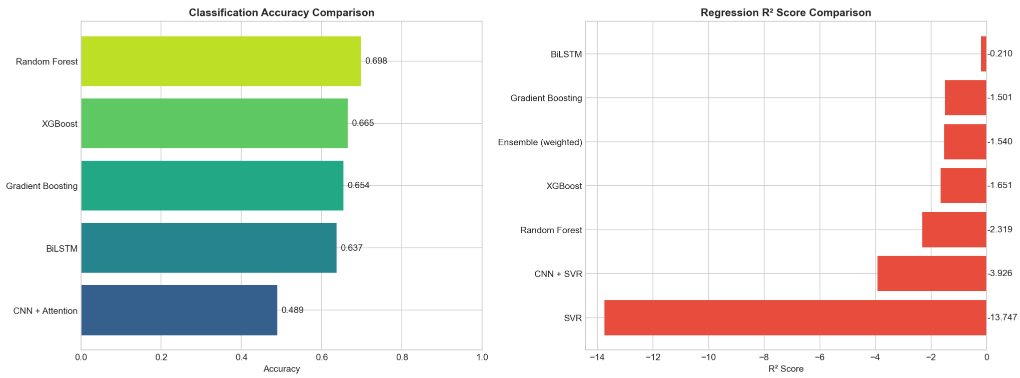

Figure 8

Comparative performance of all models across classification (accuracy, macro F1) and regression (RMSE, R^2) metrics.

Deployment Considerations

To bridge the gap between offline model evaluation and operational deployment, an operations dashboard prototype was developed (Figure 9). The dashboard translates model outputs into a traffic-light visualisation (green/yellow/red) corresponding to the three health states, providing production managers with an immediately actionable display without requiring interpretation of numerical RUL values. Integration with existing SCADA or MES infrastructure would require real-time feature extraction and model inference pipelines, which are beyond the scope of this study but represent a natural next step.

Conclusions

This study developed and evaluated a comprehensive predictive maintenance pipeline for rolling element bearings, yielding the following principal findings:

1. A complete CRISP-DM pipeline was implemented, processing 7 534 vibration recordings from six PRONOSTIA bearings through noise filtering, feature extraction (34 features via lambda functions), health index construction, classification, regression, and explainability analysis.

2. For health-state classification, Random Forest achieved the highest accuracy (69.8%) and XGBoost the highest macro F1 (0.537). Deep learning models underperformed due to

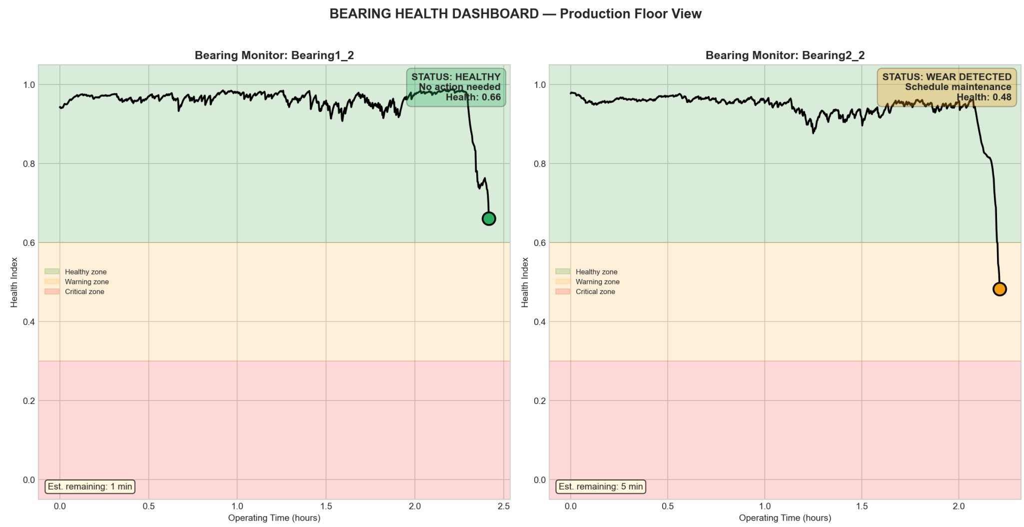

Figure 9

Operations dashboard prototype displaying bearing health status in a traffic-light format suitable for production floor use.

insufficient training data.

3. For continuous RUL regression, CNN+LSTM was the clear best performer (RMSE = $211.0 \approx 35\,\text{min}$, $R^2 = 0.241$)—the only model to achieve a positive R^2 on cross-bearing evaluation. This confirms that hybrid CNN+LSTM architectures are state-of-the-art for bearing RUL prediction, consistent with Li et al. (2018) and Zhao et al. (2019).

4. SHAP analysis confirmed RMS vibration amplitude and spectral energy as the most important predictive features, aligning with the domain knowledge established by Sutrisno et al. (2012) and ISO vibration severity standards.

5. Two distinct degradation modes were identified—gradual onset ($\sim$55% life) and sudden failure ($\sim$95%+ life)—highlighting the need for models robust to heterogeneous failure mechanisms.

6. CNN+LSTM achieved a positive R^2 (0.241) on cross-bearing evaluation, a significant result given that all other models produced negative R^2. The system provides operationally useful maintenance warnings with an average prediction error of approximately 35 minutes.

Future work should explore domain adaptation techniques to improve cross-condition generalisation, larger bearing datasets to unlock the full potential of deep learning architectures, and conformal prediction methods for better-calibrated uncertainty estimates.

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